Assignment-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

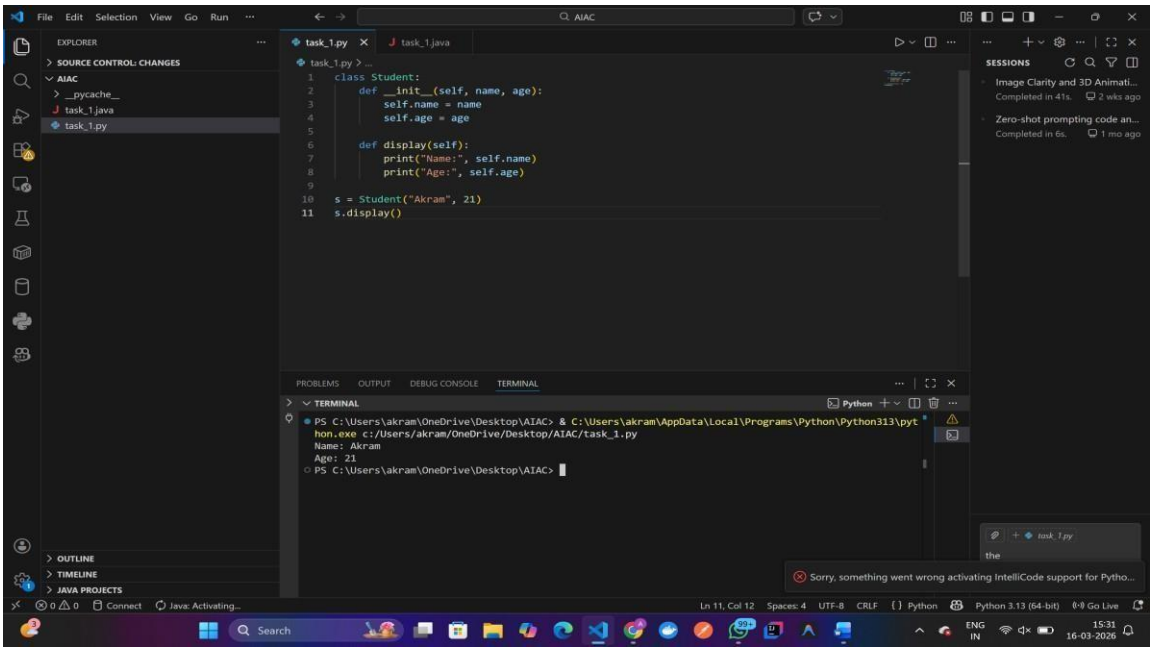
Instructions:

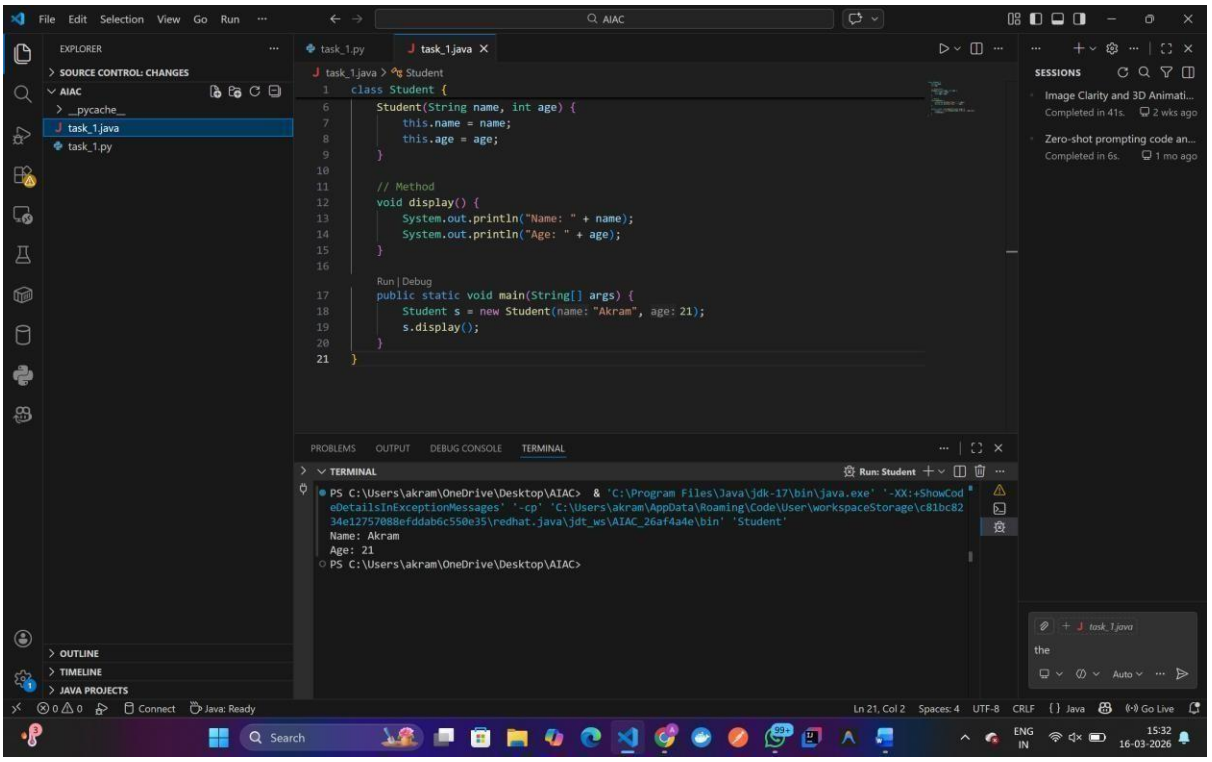
- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

CODE:





Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

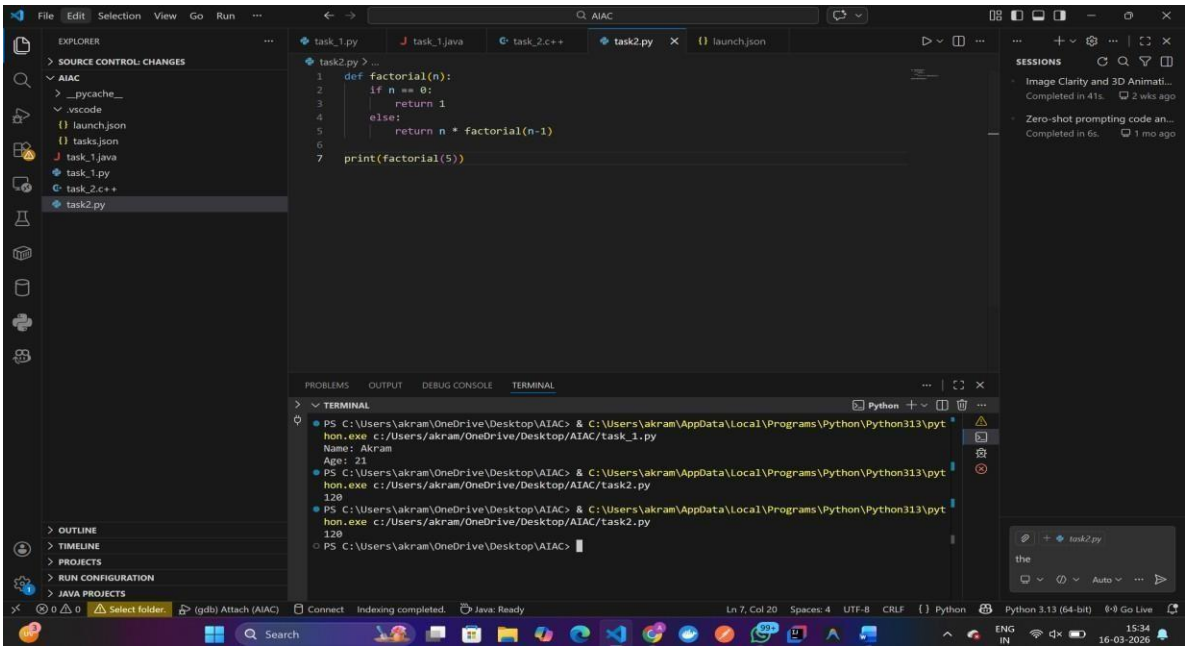
Instructions:

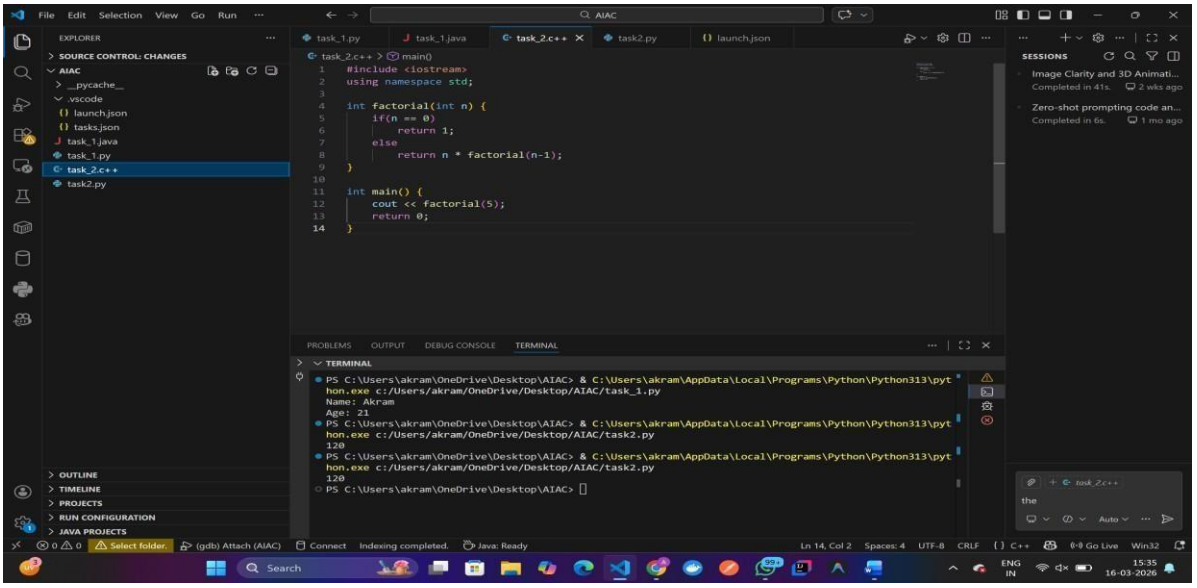
- Ask AI to translate a recursive C++ factorial function into Python.
- Validate correctness by testing with sample inputs.
- Ensure Python version follows Pythonic conventions.

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

CODE:





Task 3 – SQL to Pandas Query Task:

Translate a SQL query into a Pandas DataFrame operation in

Python.

Instructions:

• Provide AI with a SQL query (e.g., SELECT name, salary

FROM employees WHERE salary > 50000).

• Ask AI to generate equivalent Pandas code.

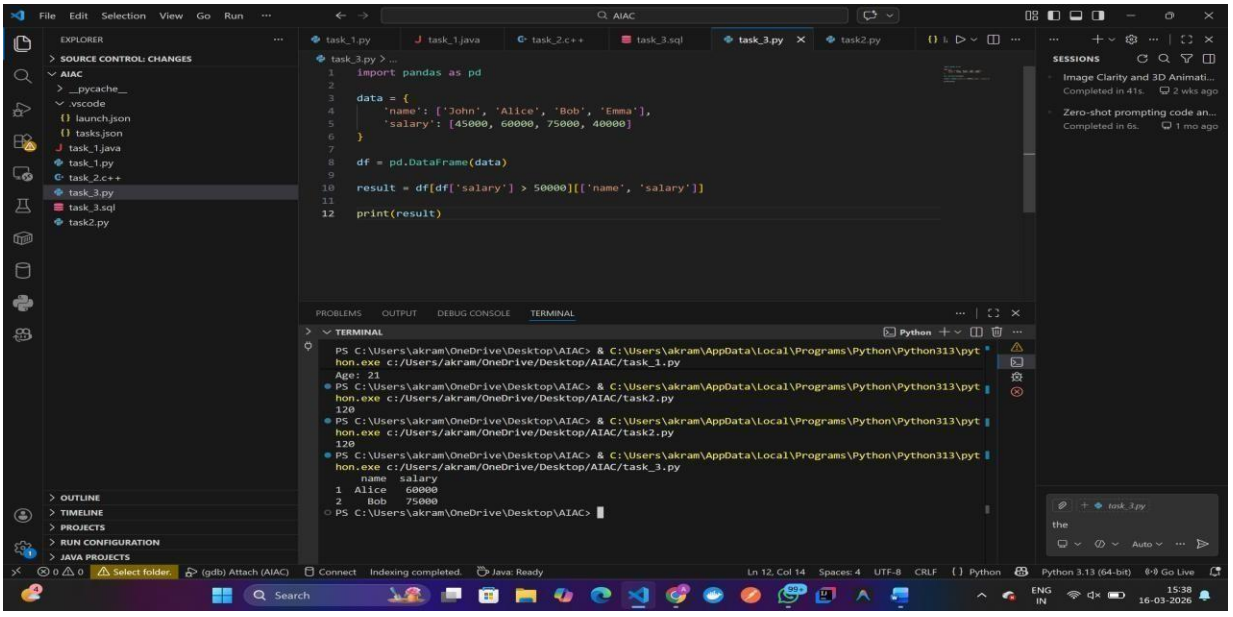
• Test the generated code on a sample DataFrame.

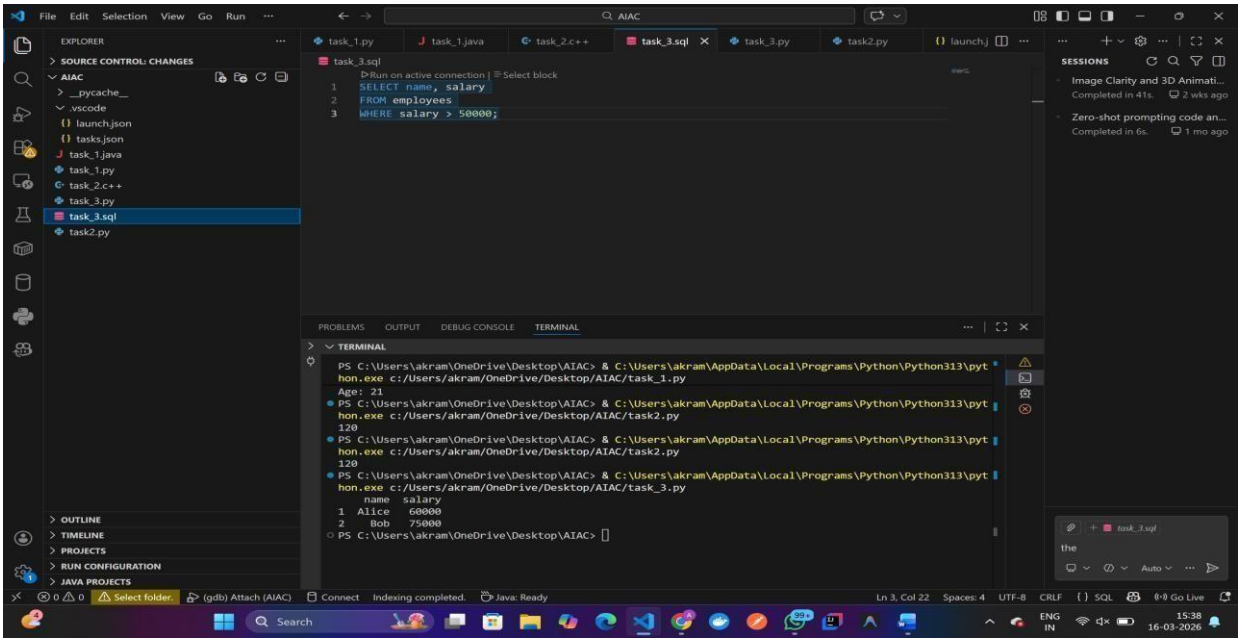
Expected Output:

• Equivalent Pandas query that retrieves the correct subset of

data.

CODE:





Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

- Converting a Python web scraping snippet into JavaScript for browser automation.
- Translating a Java method into C# for enterprise applications.

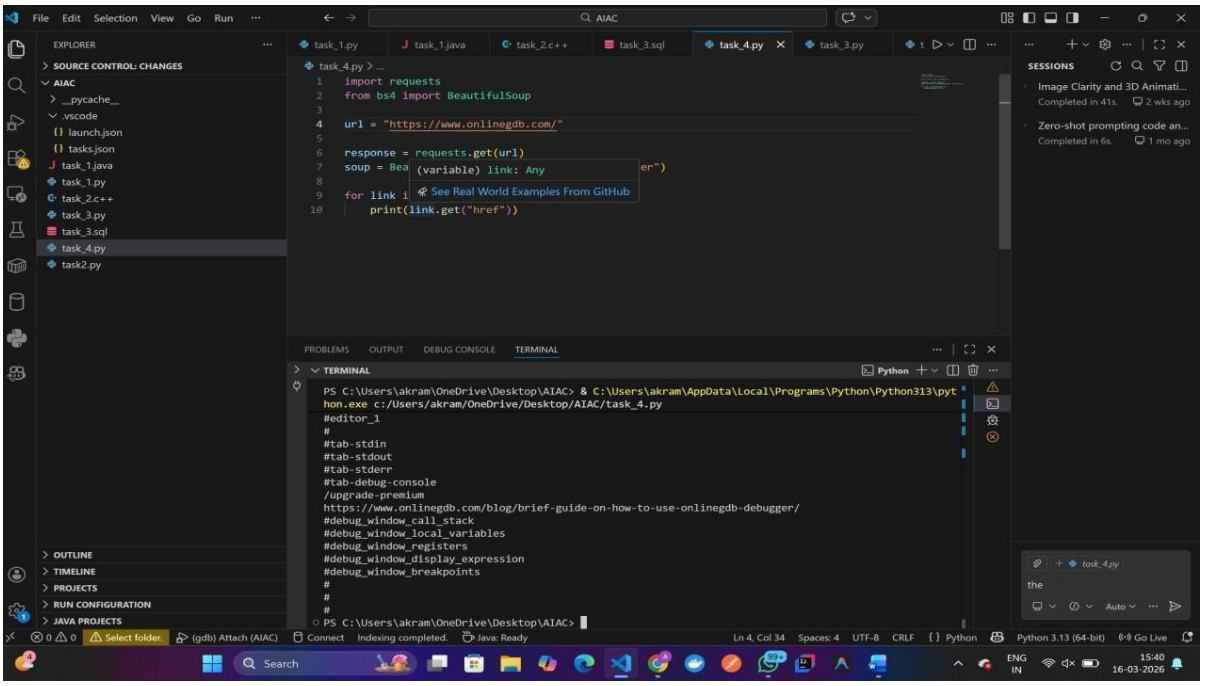
Instructions:

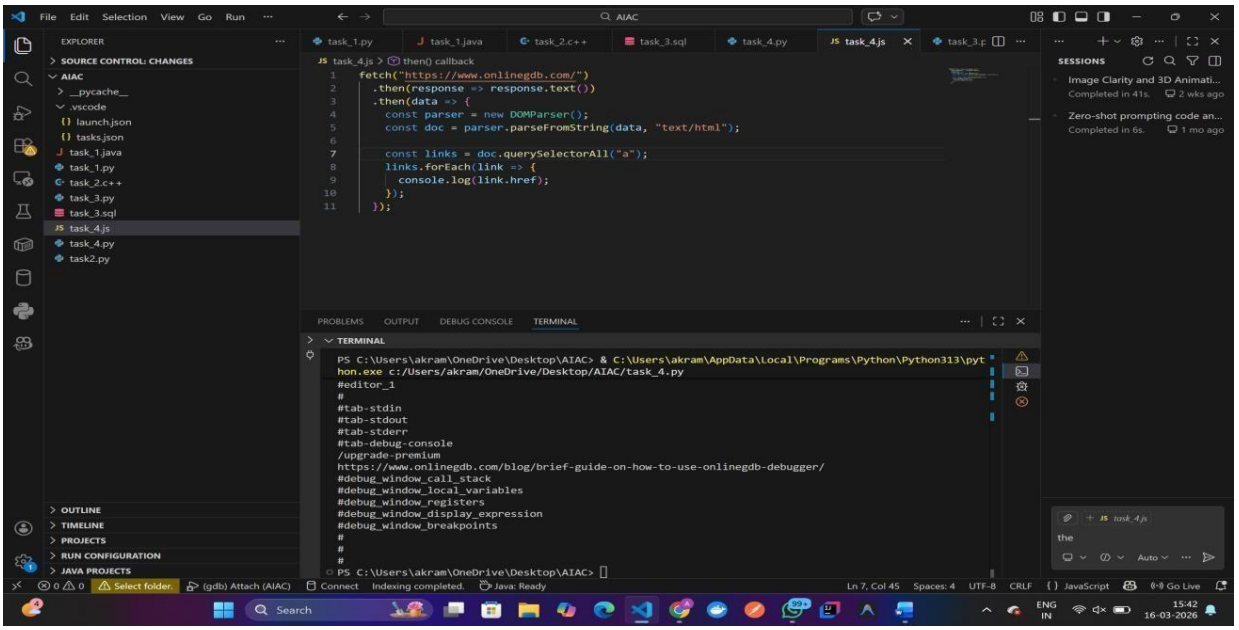
- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

- Working equivalent code snippets in at least two different programming languages

CODE:




File Edit Selection View Go Run
AIAC
EXPLORER
SOURCE CONTROL: CHANGES
AIAC
__pycache__
.vscode
launch.json
tasks.json
task_1.java
task_1.py
task_2.c++
task_3.py
task_3.sql
task_4.js
task_4.py
task2.py
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
task_1.py
task_1.java
task_2.c++
task_3.sql
task_4.py
task_4.js
task_4.js > then() callback
fetch("https://www.onlinegdb.com/")
.then(response => response.text())
.then(data => {
const parser = new DOMParser();
const doc = parser.parseFromString(data, "text/html");
const links = doc.querySelectorAll("a");
links.forEach(link => {
console.log(link.href);
});
});
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
TERMINAL
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_4.py
#editor_1
#
#tab-stdin
#tab-stdout
#tab-stderr
#tab-debug-console
/upgrade-premium
https://www.onlinegdb.com/blog/brief-guide-on-how-to-use-onlinegdb-debugger/
#debug_window_call_stack
#debug_window_local_variables
#debug_window_registers
#debug_window_display_expression
#debug_window_breakpoints
#
#
#
PS C:\Users\akram\OneDrive\Desktop\AIAC>
SESSIONS
Image Clarity and 3D Animati...
Completed in 41s. 2 wks ago
Zero-shot prompting code an...
Completed in 6s. 1 mo ago
task_4.js
the
Auto
Select folder.
(gdb) Attach (AIAC)
Connect
Indexing completed.
Java: Ready
Ln 7, Col 45
Spaces: 4
UTF-8
CRLF
JavaScript
Go Live
Search
ENG IN
15:42
16-03-2026